

— STATISTICAL INFERENCE PROJECT

Penguin morphometrics

Body mass modelling and species
classification on 324 complete
observations from the Palmer Archipelago.

Linear regression

Two-way ANOVA

Logistic regression

Antonio Arbizzani Tonelli · Simone Biraghi · Samuele Cipriani ·
Tommaso Frignani



Research questions

The same four morphological measurements support two analyses that answer different questions.

01

What explains the variation in body mass?

A continuous response measured in grams, with morphological and categorical predictors.

Multiple linear regression

slides 05–11

Two-way ANOVA

slides 13–14

02

Can measurements separate Adelie from Chinstrap?

A binary response on the 206 penguins of the two species that overlap in size.

Logistic regression

slides 16–17

Gentoo is excluded: it separates trivially

Dataset and variables

324

complete observations, one row per penguin,
after dropping every record with a missing field

RESPONSE

Body mass (g)

MEASURED IN MILLIMETRES

Bill length · Bill depth · Flipper length

CATEGORICAL

Species · Sex · Island · Clutch completion

COMPOSITION OF THE SAMPLE

Species 324 penguins



Island 324 penguins



Sex 324 penguins



Clutch completion 324 penguins



Mean body mass is 3703 g for Adelie, 3730 g for Chinstrap and 5091 g for Gentoo. Adelie and Chinstrap are almost indistinguishable on mass alone, which is exactly why the classifier uses shape instead.

Modelling body mass

From the strongest univariate association to a multivariable model, its diagnostics, its simplification and its out-of-sample error.

05
Correlation

06
Specification

07
Fit

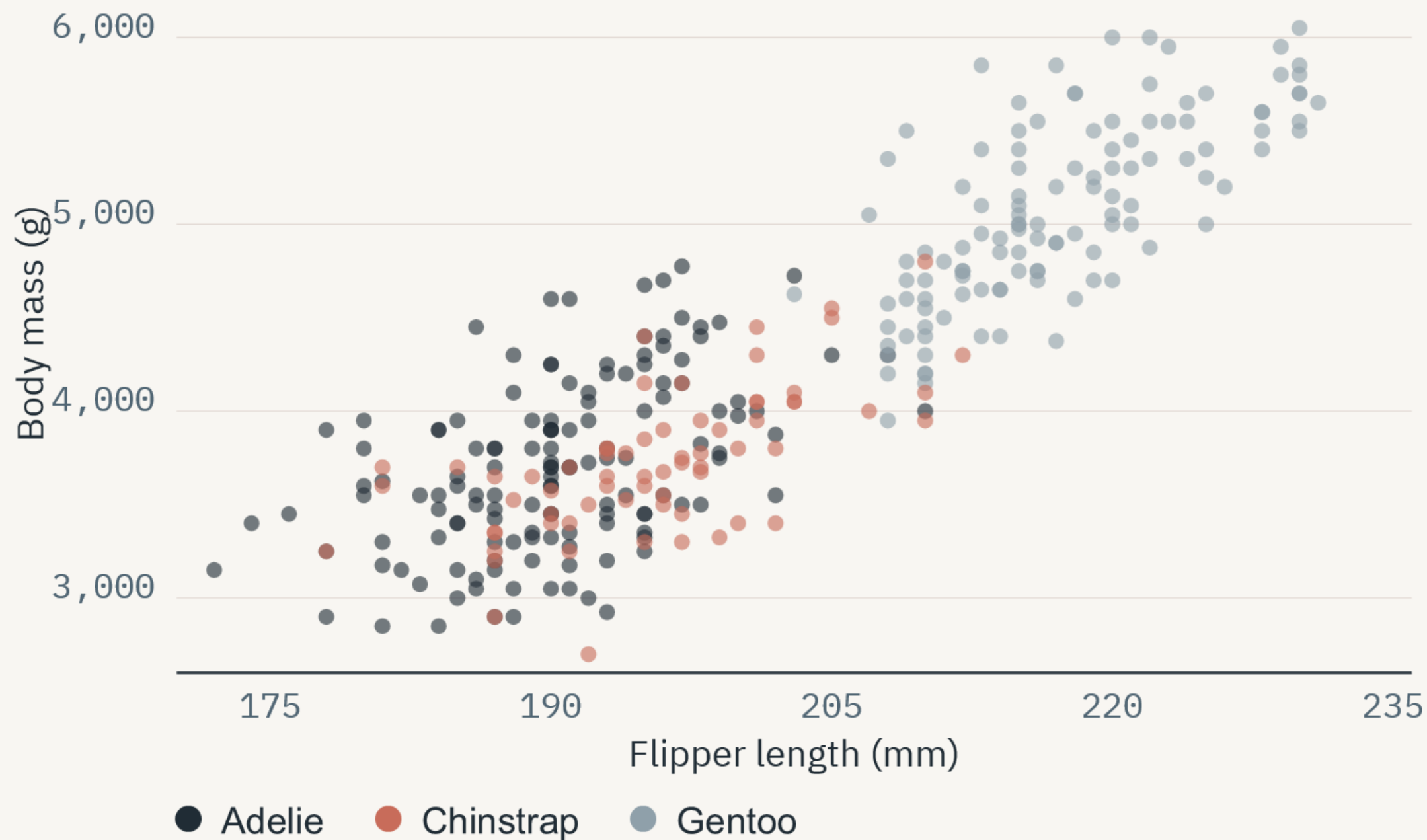
08
Diagnostics

09
Selection

10
Intervals

11
Validation

Flipper length carries most of the univariate signal



0.88

Pearson correlation between flipper length and body mass, $n = 324$

The negative bill-depth correlation is a between-species effect, not a biological rule: Gentoo are the heaviest birds and also the shallowest-billed. Pooled correlations therefore cannot be read as within-species relationships.

Multivariable body-mass model

```
body mass ~ species + island + clutch
completion + bill length + bill depth
+ flipper length + sex
```

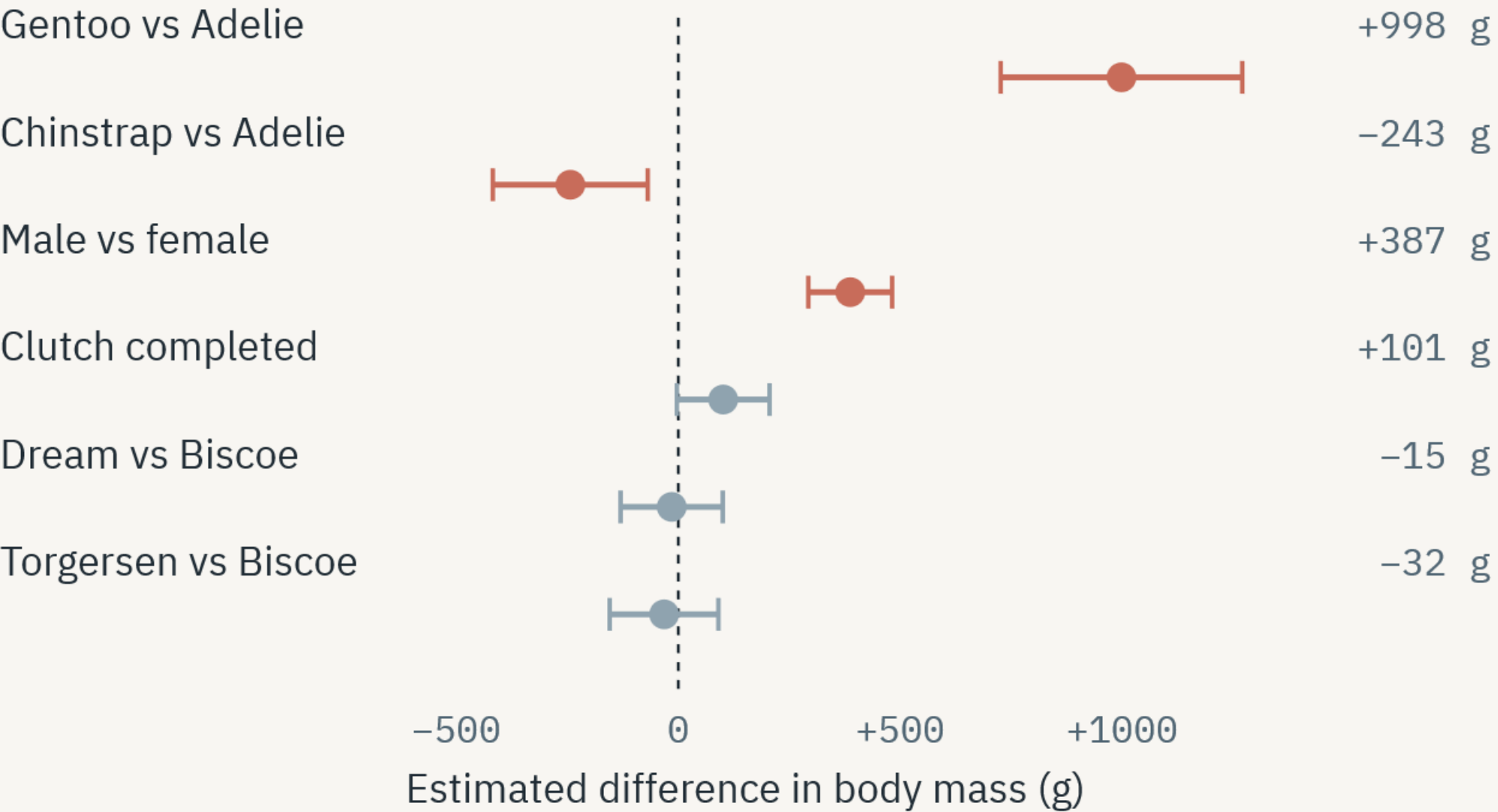
REFERENCE LEVELS

Adelie · Biscoe · clutch not completed · female

CONTINUOUS SLOPES, GRAMS PER MM

Flipper length	+16.0	[10.1, 21.8]
Bill depth	+69.3	[30.3, 108.3]
Bill length	+18.9	[4.9, 32.9]

Categorical contrasts in grams, with 95% intervals



Nine predictors explain 88% of the variation, and typical error is a quarter of a kilogram

0.876

adjusted R^2 , from $R^2 = 0.880$ on 9 and 314 degrees of freedom

285 g

residual standard error, about 7% of the mean body mass

255.3

F statistic on 9 and 314 df, $p < 2.2 \times 10^{-16}$

WHAT THE COEFFICIENTS SAY

A Gentoo weighs **998 g** more than an Adelie of the same size and sex, and a male weighs **387 g** more than a female of the same species. Both contrasts hold the morphology fixed.

Each millimetre of flipper adds **16.0 g**, and the bill measurements add **18.9 g** and **69.3 g** per mm — bill depth with the opposite sign to its pooled correlation.

WHAT THEY DO NOT SAY

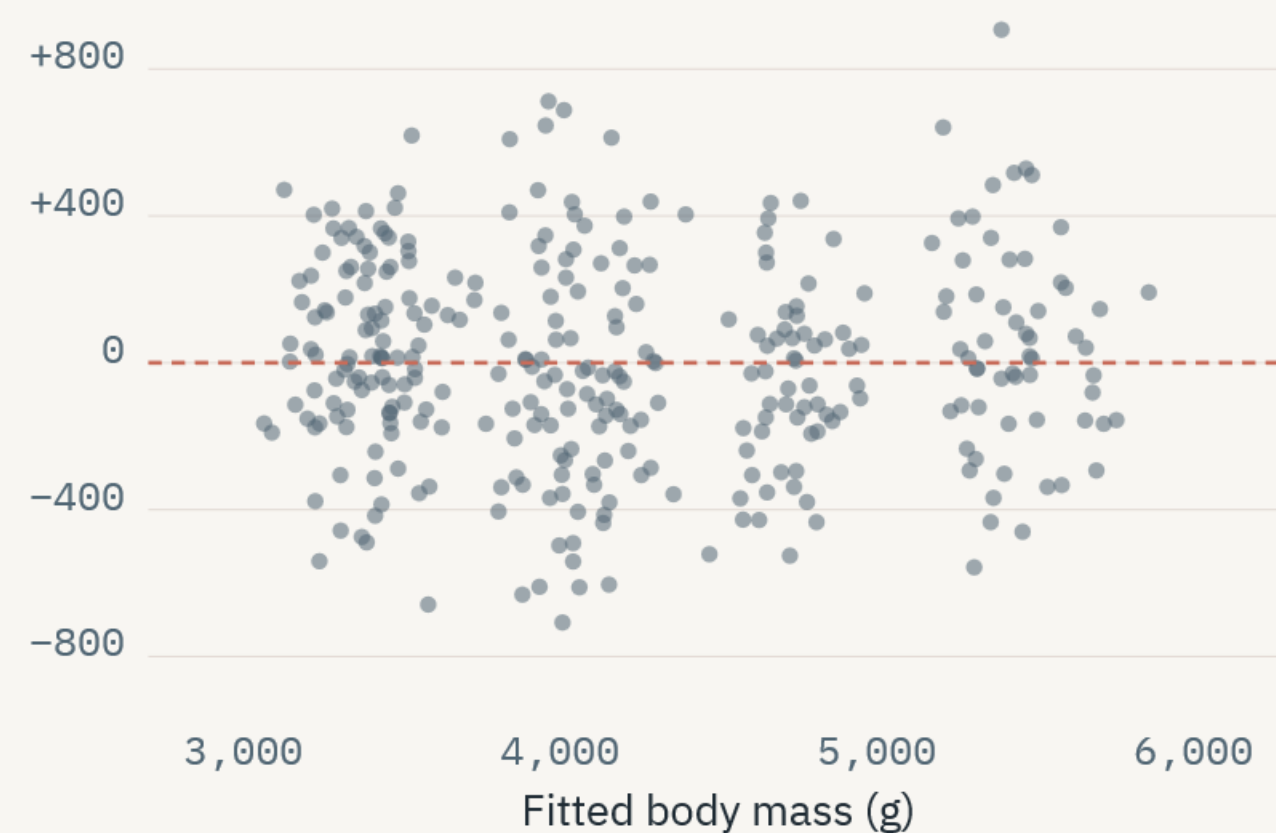
After adjustment, the island contrasts are small and imprecise: Dream -15 g ($p = 0.80$) and Torgersen -32 g ($p = 0.61$). Clutch completion is borderline at $+101$ g ($p = 0.058$).

The design is observational, so these are associations within the sample. Nothing here licenses a causal reading of flipper length on mass.

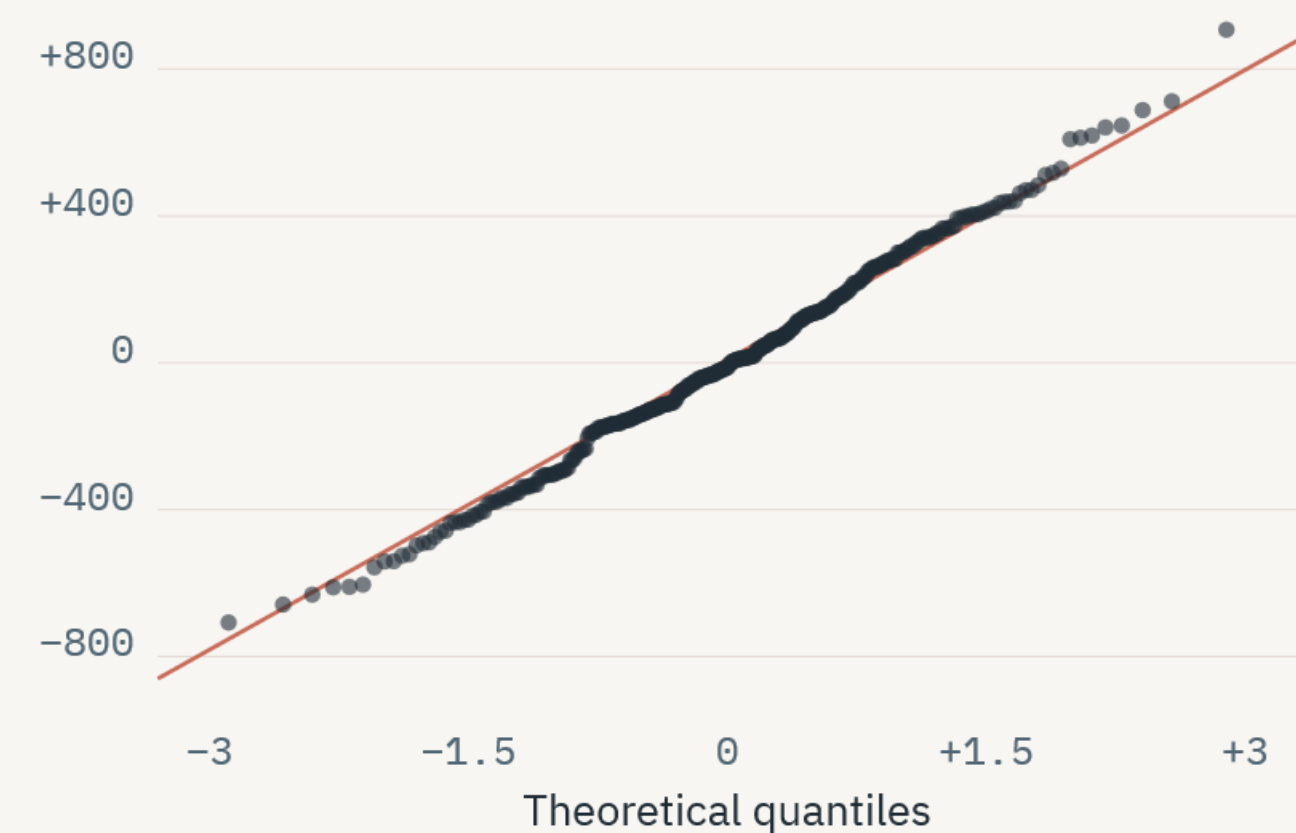
The assumptions hold well enough to keep the model as it is

No curvature and no fan shape in the residuals; the tails of the Q-Q plot are slightly heavy, which is what we expect with 324 observations and no transformation.

Residuals versus fitted values



Normal Q-Q plot of residuals



29 observations are flagged by at least one criterion: 11 high leverage, 14 with a standardised residual beyond ± 2 , 18 above the $4/(n - p)$ cut for Cook's distance. The largest Cook's distance is **0.037**, far from 1, so no single bird drives the fit.

Dropping island and clutch completion costs nothing

Backward elimination with penalty $k = \log n$ removes the two blocks that the coefficient plot already showed as flat.

MODEL	TERMS	BIC	AIC	ADJ. R ²
Full model	7	4634.8	4593.2	0.8763
BIC-selected model	5	4621.8	4591.5	0.8759

RETAINED

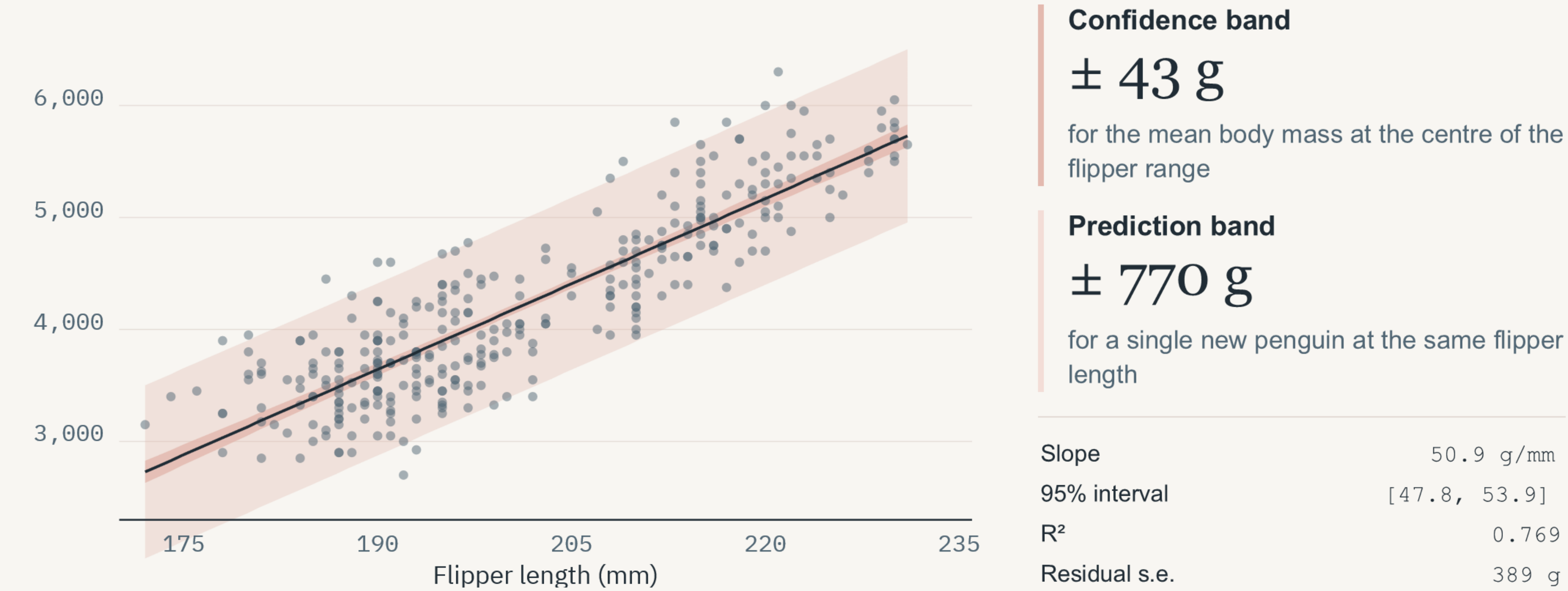
species + bill length + bill depth + flipper
length + sex

REMOVED

island + clutch completion

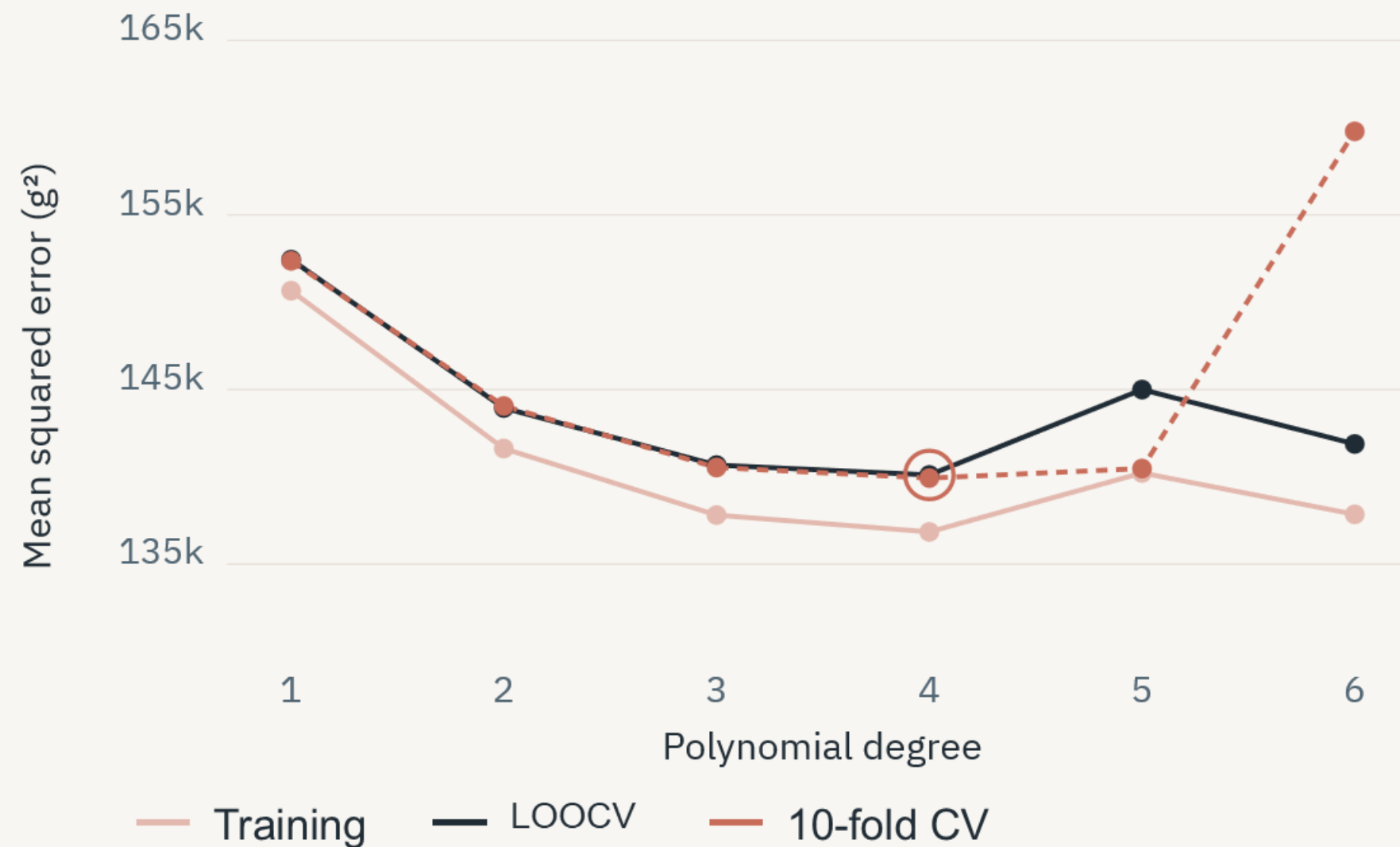
Adjusted R^2 moves by 0.0004 and the residual standard error by half a gram, from 284.6 g to 285.1 g. The simplification is a statement about parsimony; it is not evidence that island has no effect, only that it adds nothing beyond species.

Predicting the average penguin is 18 times more precise than predicting one penguin



Added curvature buys 16 grams of accuracy, so we keep the straight line

Out-of-sample error for polynomials in flipper length. Cross-validated error falls from degree 1 to degree 4 and then breaks down.



Degree 1, LOOCV

390 g

Degree 4, LOOCV

374 g

A 4% gain in root mean squared error, against coefficients we could no longer interpret. Beyond degree 4 the raw polynomial design is ill-conditioned and the two estimates disagree by more than the gain itself.

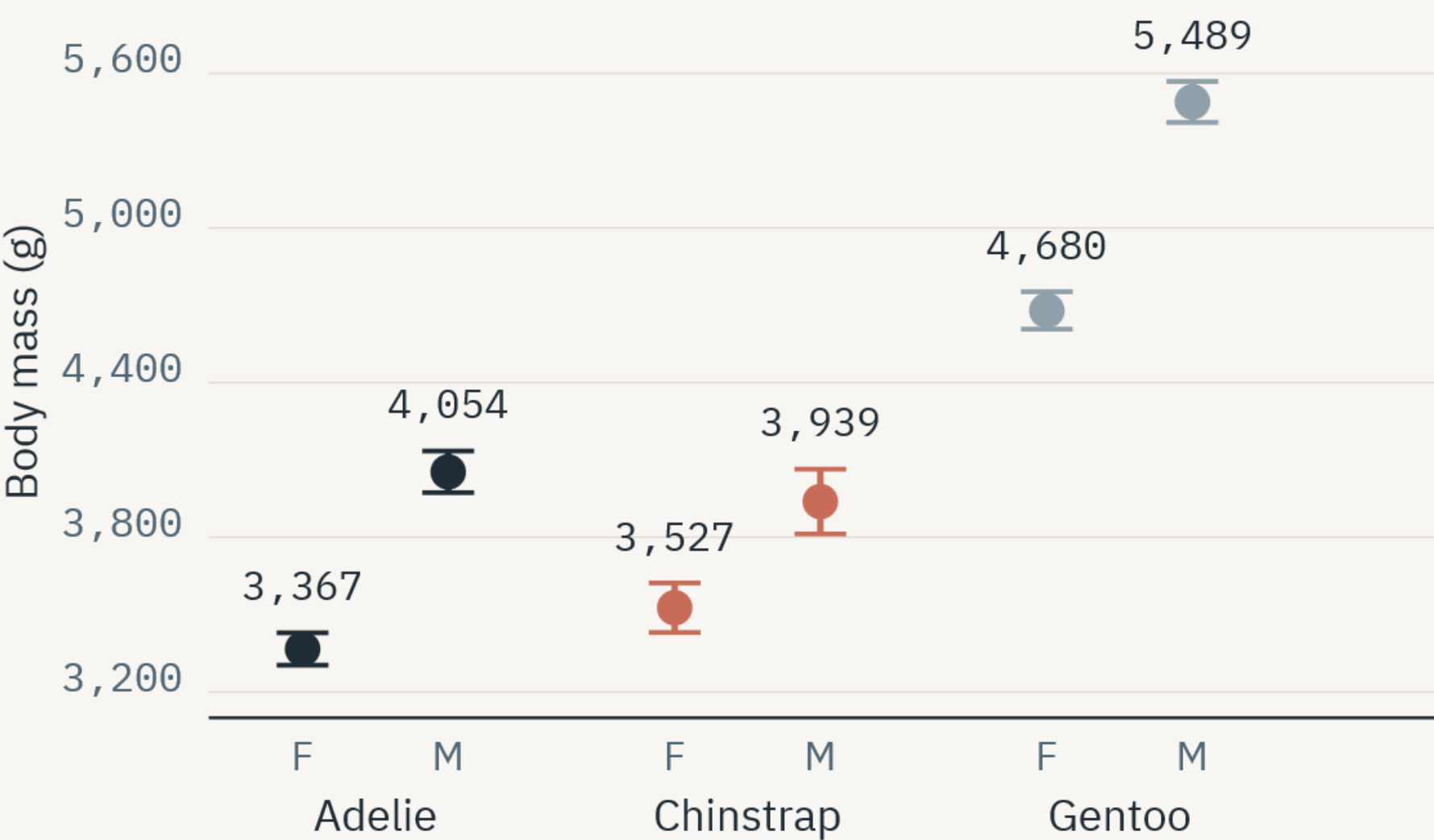
Curvature is not what limits this model: adding species and sex is worth far more (appendix A3).

Species and sex

A two-way ANOVA on body mass, and what a significant interaction term actually means for the size difference between males and females.

Species and sex both matter, and they do not act independently

Mean body mass by species and sex, with 95% intervals

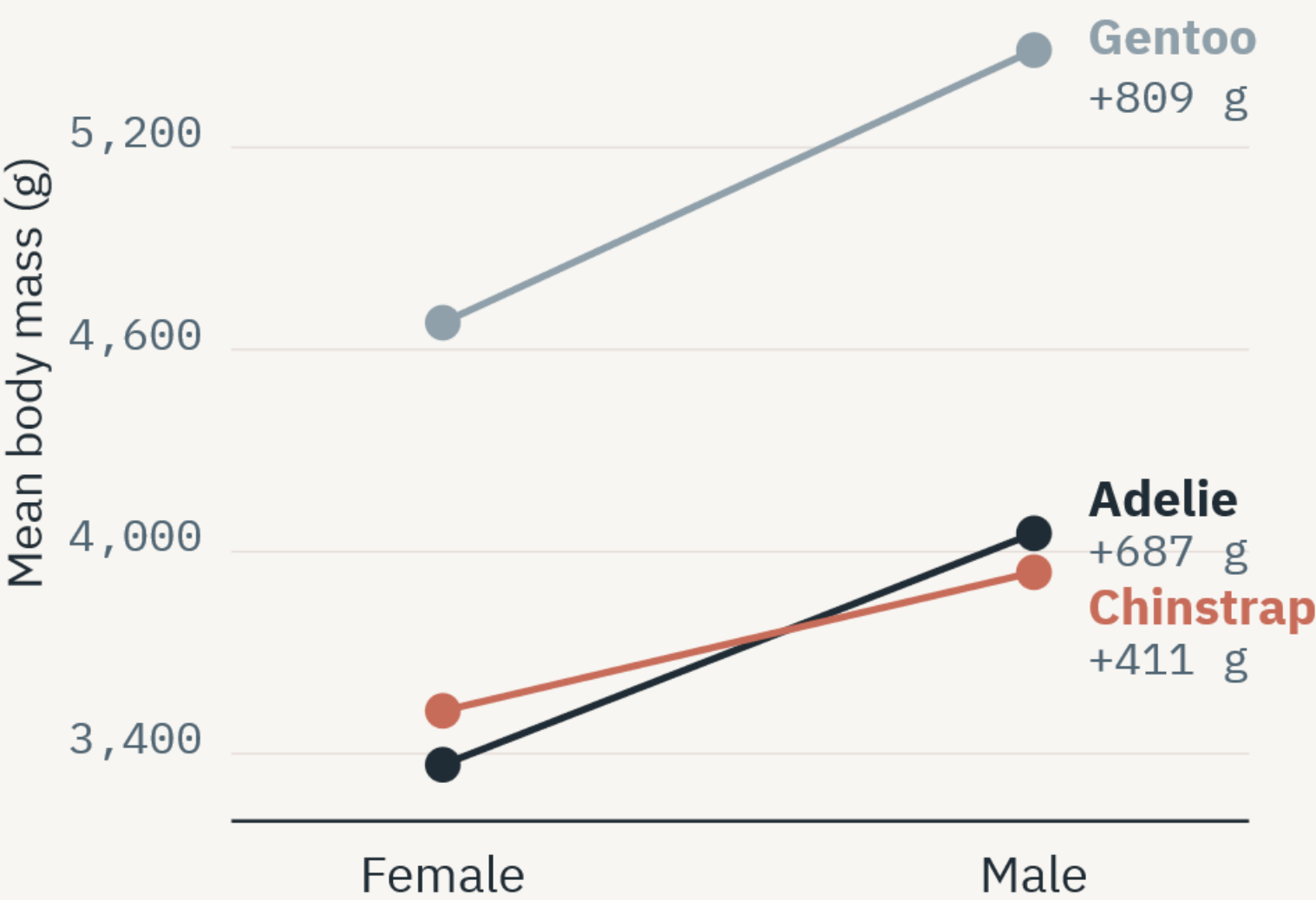


SOURCE	DF	F	P
Species	2	752.3	<2e-16
Sex	1	388.1	<2e-16
Species × sex	2	8.94	0.000166
Residuals	318	—	—

Sequential sums of squares with species entered first.
Residual mean square 94 919 g², i.e. a within-group standard deviation of about 308 g.

Chinstrap sexual dimorphism is half that of Gentoo

The interaction is not about whether males are heavier — they always are — but about how much heavier, which changes with species.



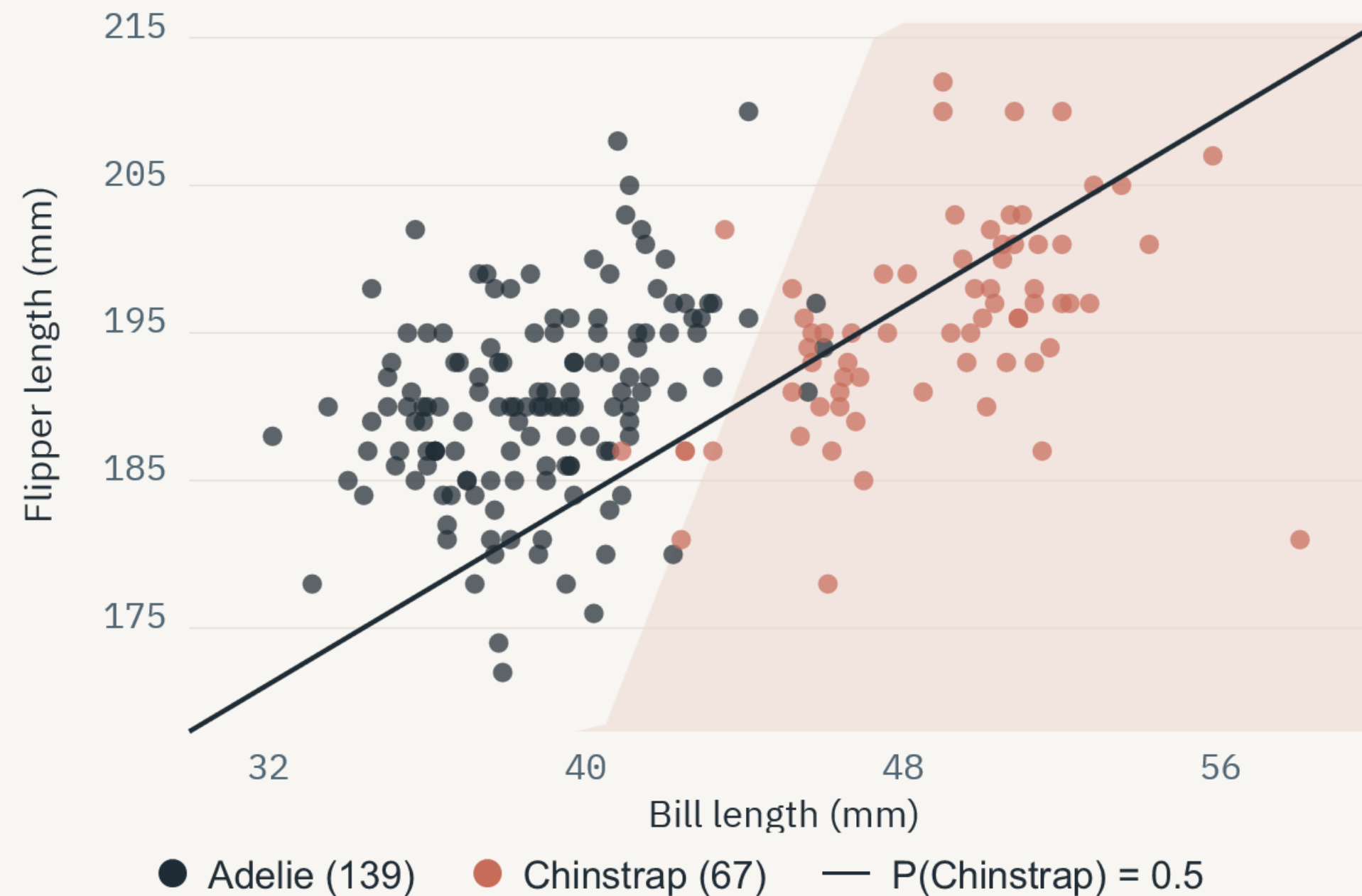
MALE MINUS FEMALE, TUKEY 95% INTERVALS		
Adelie	+687 g	20.4% of female mass
Chinstrap	+411 g	11.7% of female mass
Gentoo	+809 g	17.3% of female mass

All three gaps exclude zero, so dimorphism is real in every species. What the interaction adds is that the Chinstrap gap is materially smaller — the reason the three lines are not parallel.

Telling Adelie from Chinstrap

Two measurements, one decision boundary, and an evaluation that we deliberately do not oversell.

Bill length separates the two species; flipper length corrects for body size



$$\text{logit } P(\text{Chinstrap}) = -19.59 + 1.223 \cdot \text{bill length} - 0.178 \cdot \text{flipper length}$$

Odds ratio, bill length	3.40 ×
95% interval	[2.19, 5.27]
Odds ratio, flipper	0.84 ×
95% interval	[0.73, 0.97]

One extra millimetre of bill multiplies the odds of being a Chinstrap by 3.4 at fixed flipper length. Adding flipper length to the bill-only model is worth a likelihood-ratio χ^2 of 7.11 on 1 df ($p = 0.008$).

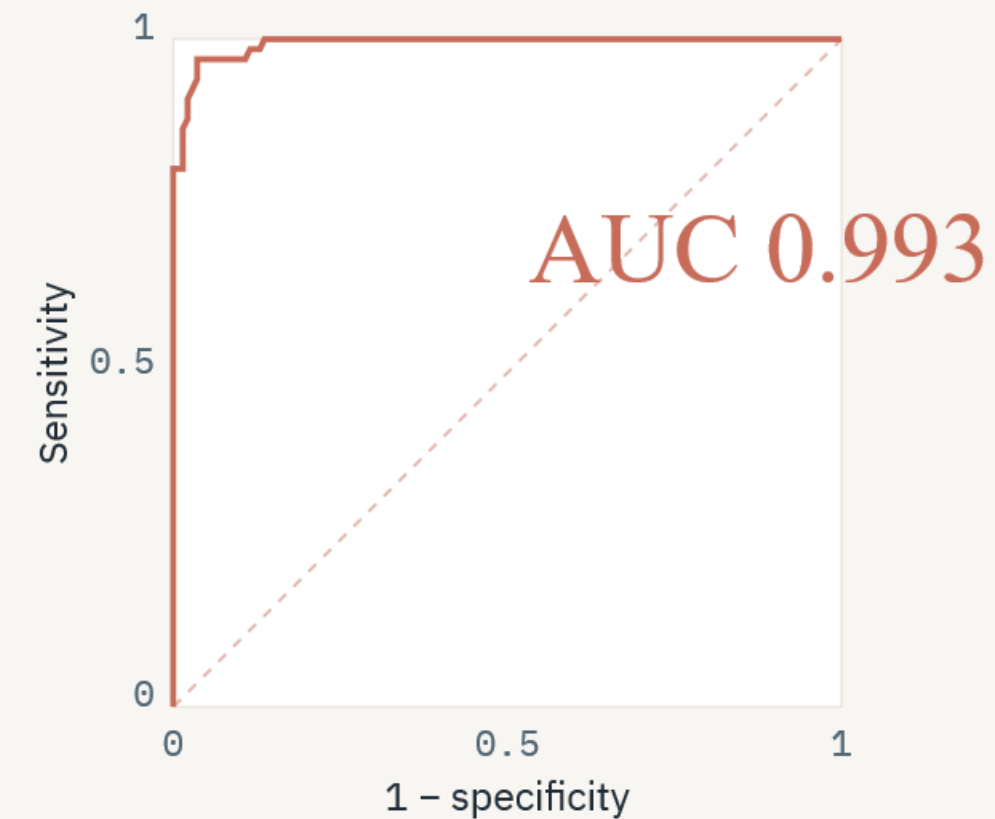
Eight birds out of 206 fall on the wrong side of the line

All figures below are in-sample: the classifier was scored on the same observations used to estimate it.

CONFUSION MATRIX, THRESHOLD 0.5

	Predicted Adelie	Predicted Chinstrap
Observed Adelie	135	4
Observed Chinstrap	4	63
96.1% accuracy	94.0% sensitivity	97.1% specificity

ROC CURVE



An AUC of 0.993 says the two species are almost linearly separable in this plane. It does not say the classifier would hold on penguins measured by another team, on another island, in another season.

What these results cannot establish

01

The design is observational

No measurement was assigned, so the coefficients are conditional means, not effects. We can say heavier birds have longer flippers; we cannot say a longer flipper makes a bird heavier.

03

The results are model-dependent

Bill depth changes sign between the pooled correlation and the fitted model. Any single coefficient has to be quoted together with the set of variables it was conditioned on.

02

Complete cases only

We dropped 20 of 344 records with a missing field. If those measurements failed for a reason connected to the bird's condition, our sample is mildly selected and the means are biased.

04

The classifier is not validated

96.1% and AUC 0.993 are in-sample. The size of the optimism is unknown here. An external estimate would require nested resampling or an independent test set.

Three answers

16.0 g

per millimetre of flipper, species and sex held fixed

Flipper length is the strongest continuous predictor of body mass, and the multivariable model accounts for 88% of the adjusted variation with a residual standard error of 285 g. BIC removes island and clutch completion after accounting for species and morphology.

$p = 1.7e-4$

species × sex interaction

Species and sex both shift body mass, and their interaction is significant: the male–female gap runs from **411 g** in Chinstrap to **809 g** in Gentoo. A single additive dimorphism constant would misdescribe the data.

96.1%

in-sample accuracy, AUC 0.993

Bill length and flipper length separate Adelie from Chinstrap almost linearly within this sample. The honest next step is not a bigger model but an independent test set.

Appendix

Full regression table, influence diagnostics, the value of each added predictor, and the goodness-of-fit checks for the classifier.

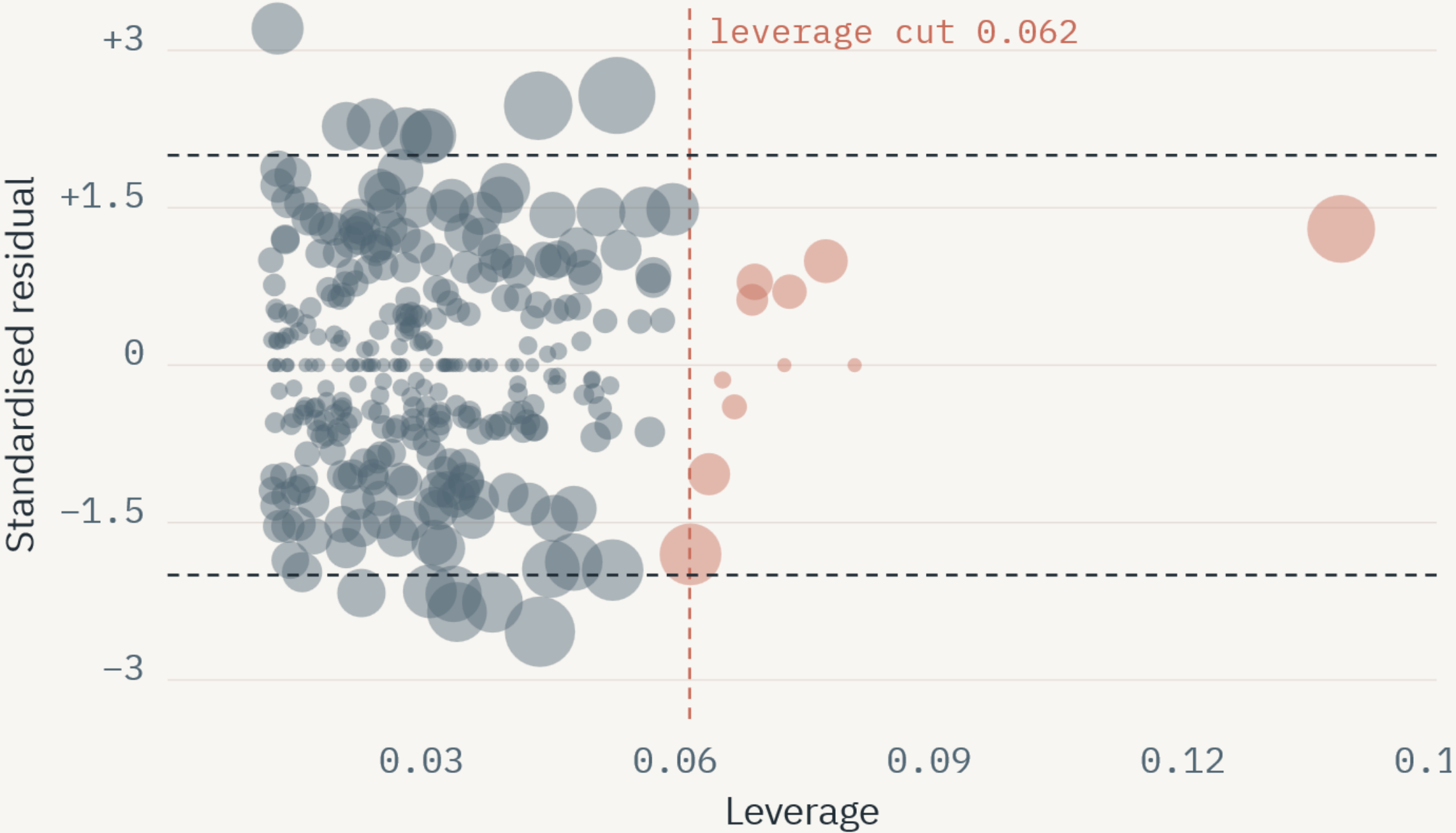
Full linear model, complete coefficient table

TERM	ESTIMATE	STD. ERROR	95% INTERVAL	P VALUE
Intercept	-1603.5	579.2	[-2739, -468]	0.006
Chinstrap vs Adelie	-243.2	89.0	[-418, -69]	0.007
Gentoo vs Adelie	997.7	138.7	[726, 1270]	4.7e-12
Dream vs Biscoe	-14.9	58.7	[-130, 100]	0.800
Torgersen vs Biscoe	-32.0	62.4	[-154, 90]	0.609
Clutch completed	101.1	53.2	[-3, 205]	0.058
Bill length (mm)	18.9	7.1	[5, 33]	0.008
Bill depth (mm)	69.3	19.9	[30, 108]	5.6e-4
Flipper length (mm)	16.0	3.0	[10, 22]	2.1e-7
Male vs female	387.1	48.2	[293, 482]	2.0e-14

Reference levels: Adelie, Biscoe, clutch not completed, female. Residual standard error 284.6 g on 314 degrees of freedom; $R^2 = 0.880$, adjusted $R^2 = 0.876$; $F(9, 314) = 255.3$.

Influence diagnostics for the full model

Bubble area is proportional to Cook's distance

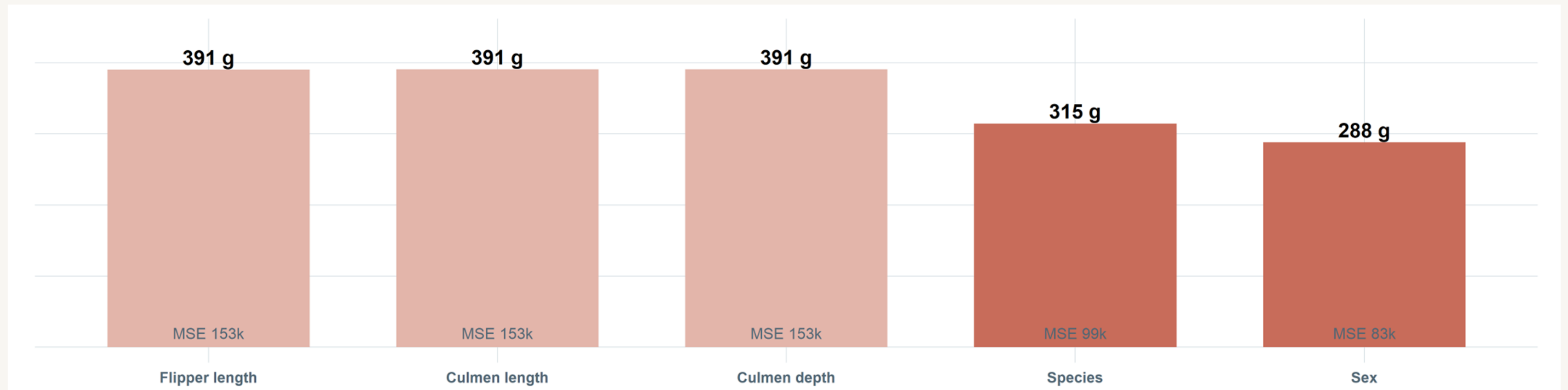


High leverage, $> 2p/n$	11
$ \text{standardised residual} > 2$	14
Cook's $D > 4/(n - p)$	18
Flagged by any criterion	29
Largest Cook's D	0.037

The $4/(n - p)$ rule is deliberately sensitive; with 324 observations the cut-off sits at 0.013. Judged on the conventional threshold of 1, no observation is influential, so we refit nothing and delete nothing.

Species and sex drive the main reduction in prediction error

Ten-fold cross-validated root mean squared error for progressively larger models.



Adding the two bill measurements to a flipper-only model does not reduce cross-validated error. In this sequence, species lowers RMSE from 391 to 315 g, and sex lowers it further to 288 g. This complements the BIC comparison, but answers a different out-of-sample question.

Classifier: fit, comparison and goodness of fit

DEVIANCE, N = 206

Null deviance, 205 df	259.9
Bill length only, 204 df	50.0
Bill + flipper, 203 df	42.9
Likelihood-ratio χ^2 , 1 df	7.11 (p = 0.008)

GOODNESS OF FIT

Hosmer–Lemeshow χ^2 , 8 df	2.19
p value	0.975

WHY NOT THE FULL LOGISTIC MODEL

With island, clutch completion, all four measurements and sex, the fit runs into quasi-complete separation: Chinstrap are observed on Dream only, so the island coefficient diverges and its standard error explodes. The likelihood keeps increasing without a finite maximum.

We keep the two-variable model because it converges, it can be plotted, and its coefficients can be read as odds ratios.

The Hosmer–Lemeshow test does not reject the calibration of the bivariate model, but with 8 misclassified birds out of 206 the test has very little power here. We report it for completeness, not as evidence of good calibration.